Geometric Sequences

Algebra Worksheet · Grade 9–11

Name: _____ Date: _____

Learning Objectives

- Identify whether a sequence is geometric by finding a common ratio
- Apply the geometric sequence formula to find specific terms
- Solve for unknown values such as the first term, common ratio, or term number

Problems

1. Find the common ratio of the geometric sequence below.

$2,\ 6,\ 18,\ 54$

2. Determine whether the following sequence is a geometric sequence. Explain why or why not.

$4,\ 8,\ 16,\ 32$

3. Determine whether the following sequence is a geometric sequence. Explain why or why not.

$3,\ 6,\ 10,\ 15$

4. Find the 6th term of the geometric sequence below using the formula a sub n = a sub 1 times r to the power of (n minus 1).

$2,\ 6,\ 18,\ 54,\ \ldots$

5. Find the 8th term of the geometric sequence below.

$5,\ 10,\ 20,\ 40,\ \ldots$

6. Find the 5th term of the geometric sequence with a first term of 64 and a common ratio of one half.

$a_1 = 64,\ r = \frac{1}{2},\ n = 5$

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7. A geometric sequence has a first term of 7 and a common ratio of 3. Find the 7th term.

$a_1 = 7,\ r = 3,\ n = 7$

8. The 1st term of a geometric sequence is 4 and the 4th term is 108. Find the common ratio.

$a_1 = 4,\ a_4 = 108$

9. Which term of the geometric sequence below equals 1458?

$2,\ 6,\ 18,\ 54,\ \ldots$

10. A ball is dropped from a height of 81 meters. Each bounce reaches two thirds of the previous height. Find the height of the 5th bounce.

$a_1 = 81,\ r = \frac{2}{3},\ n = 5$

Geometric Sequences — Answer Key

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Answer Key

1. Answer: r = 3

- Divide the second term by the first: 6 ÷ 2 = 3
- Verify: 18 ÷ 6 = 3 and 54 ÷ 18 = 3
- The common ratio is r = 3

2. Answer: Yes; r = 2

- Divide consecutive terms: 8 ÷ 4 = 2
- 16 ÷ 8 = 2 and 32 ÷ 16 = 2
- All quotients are equal, so the sequence is geometric with r = 2

3. Answer: No; no common ratio

- Divide consecutive terms: 6 ÷ 3 = 2
- 10 ÷ 6 ≈ 1.67 and 15 ÷ 10 = 1.5
- The ratios are not equal, so this is NOT a geometric sequence

4. Answer: $a_6 = 1458$

- Identify: $a_1 = 2$, r = 3, n = 6
- Apply formula: $a_6 = 2 \times 3^{(6–1)} = 2 \times 3^5$
- $3^5 = 243$
- $a_6 = 2 \times 243 = 1458$

5. Answer: $a_8 = 640$

- Identify: $a_1 = 5$, r = 2, n = 8
- Apply formula: $a_8 = 5 \times 2^{(8–1)} = 5 \times 2^7$
- $2^7 = 128$
- $a_8 = 5 \times 128 = 640$

6. Answer: $a_5 = 4$

- Apply formula: $a_5 = 64 \times (1/2)^{(5–1)} = 64 \times (1/2)^4$
- $(1/2)^4 = 1/16$
- $a_5 = 64 \times (1/16) = 4$

7. Answer: $a_7 = 5103$

- Apply formula: $a_7 = 7 \times 3^{(7–1)} = 7 \times 3^6$
- $3^6 = 729$
- $a_7 = 7 \times 729 = 5103$

8. Answer: r = 3

- Use formula: $a_4 = a_1 \times r^{(4–1)}$

- 108 = 4 × r^3
- r^3 = 108 ÷ 4 = 27
- r = $\sqrt[3]{27}$ = 3

9. Answer: n = 7 (the 7th term)

- Identify: a_1 = 2, r = 3, a_n = 1458
- Set up equation: 1458 = 2 × 3^(n–1)
- Divide both sides by 2: 3^(n–1) = 729
- Since 3^6 = 729, then n – 1 = 6, so n = 7

10. Answer: a_5 = 16 meters

- Apply formula: a_5 = 81 × (2/3)^(5–1) = 81 × (2/3)^4
- (2/3)^4 = 16/81
- a_5 = 81 × (16/81) = 16
- The height of the 5th bounce is 16 meters

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